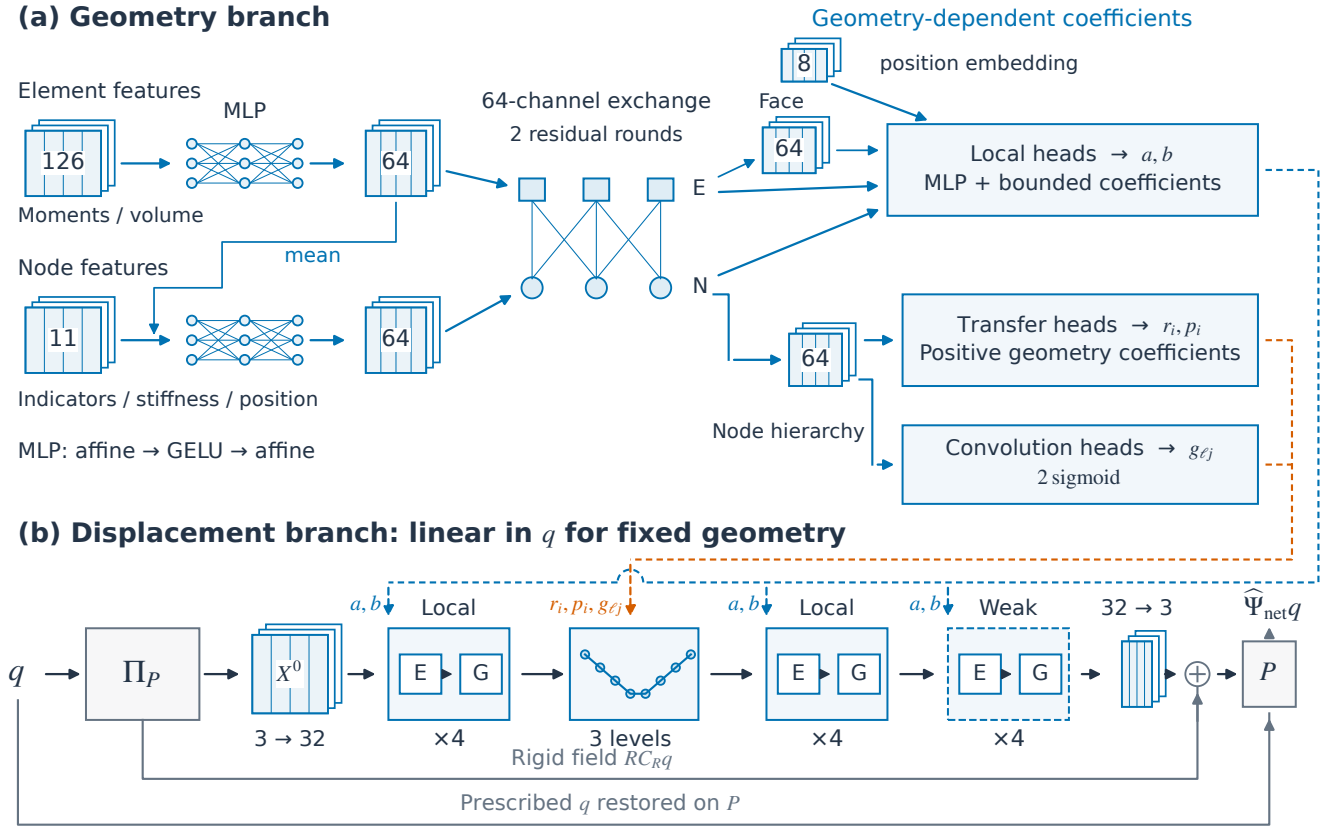

(a) Geometry branch
Geometry-dependent coefficients
Element features
MLP
126
64
Moments / volume
64-channel exchange
2 residual rounds
8
position embedding
Face
64
Local heads → a, b
MLP + bounded coefficients
E
Node features
mean
11
64
Indicators / stiffness / position
N
64
Transfer heads → r_i, p_i
Positive geometry coefficients
Node hierarchy
Convolution heads → g_ℓj
2 sigmoid
MLP: affine → GELU → affine
(b) Displacement branch: linear in q for fixed geometry
a, b
Local
r_i, p_i, g_ℓj
a, b
Local
a, b
Weak
32 → 3
Ψ̂_net q
q
Π_P
X^0
E
G
3 → 32
×4
3 levels
×4
×4
P
Rigid field RC_Rq
Prescribed q restored on P
(c) Multiscale displacement propagation
32 channels
65
33
17
9
X
X
P
R_0
σ_0 I_0
skip
2 × C
R_1
σ_1 I_1
R_2
σ_2 I_2
down ×2 → up ×2
R: restriction I: prolongation
C : X ← X + g ⊙ Conv_3×3×3(X)
(d) Local interaction
27 local nodes
4 heads
nodes
a
b
W_ℓh : 32 × 32
Restore X^0 on P
E: 27 Q2 nodes
G: 18 owner + 9 neighbour nodes
Training: update network parameters θ through both branches.
Deployment: fix θ; cache geometry coefficients.